Graph Algorithms Worksheet: Shortest Paths & TSP

Problem 1: Undirected Unweighted Graph Specification

Consider a connected, unweighted, undirected graph $G_1 = (V, E)$ consisting of **10 nodes**: $V = \{A, B, C, D, E, F, G, H, I$

Adjacency Matrix (Unweighted Edges)

An entry of **1** indicates an edge exists between the two nodes. A hyphen (**-**) indicates no direct edge exists (or self-loops).

Node	A	B	C	D	E	F	G	H	I	J
A	-	1	-	1	-	-	-	-	-	1
B	1	-	1	-	1	-	-	-	-	-
C	-	1	-	1	-	1	-	-	-	-
D	1	-	1	-	-	-	1	-	-	-
E	-	1	-	-	-	1	-	1	-	-
F	-	-	1	-	1	-	-	-	1	-
G	-	-	-	1	-	-	-	1	-	1
H	-	-	-	-	1	-	1	-	1	-
I	-	-	-	-	-	1	-	1	-	1
J	1	-	-	-	-	-	1	-	1	-

Part A: Draw the Graph

Draw the weighted graph represented by the matrix above in the workspace provided below. Label all edge weights.

Part B: Shortest Path Determination

Using the weighted adjacency matrix and your diagram, determine the **shortest path** and **total path weight** for each pair of nodes.

1. **Shortest Path from Node A to Node E:**

   Path Sequence: _____

   Total Weight: _____

2. **Shortest Path from Node C to Node H:**

   Path Sequence: _____

   Total Weight: _____

3. **Shortest Path from Node B to Node I:**

   Path Sequence: _____

   Total Weight: _____

4. **Shortest Path from Node A to Node G:**

   Path Sequence: _____

   Total Weight: _____

Part C: Travelling Salesperson Problem (TSP)

Find candidate closed tours that start and end at **Node A**, visit every node once, and minimize total weight.

Candidate Path Sequence 1: _____

Total Weight: _____

Candidate Path Sequence 2: _____

Total Weight: _____

Candidate Path Sequence 3: _____

Total Weight: _____

Problem 2: Undirected Weighted Graph Specification

Consider a connected, weighted, undirected graph $G_2 = (V, E)$ consisting of **10 nodes**: $V = \{A, B, C, D, E, F, G, H, I, J$

Adjacency Matrix (Edge Weights)

An entry at row u and column v indicates the numeric weight of edge $\{u, v\}$. A hyphen (**-**) indicates no direct edge exists.

Node	A	B	C	D	E	F	G	H	I	J
A	-	3	8	-	10	12	-	-	-	5
B	3	-	4	-	6	-	-	9	-	-
C	8	4	-	2	-	7	-	-	11	-
D	-	-	2	-	5	-	4	-	-	7
E	10	6	-	5	-	3	-	8	-	-
F	12	-	7	-	3	-	4	-	5	10
G	-	-	-	4	-	4	-	2	-	6
H	-	9	-	-	8	-	2	-	6	-
I	-	-	11	-	-	5	-	6	-	3
J	5	-	-	7	-	10	6	-	3	-

Part A: Draw the Graph

Draw the weighted graph represented by the matrix above in the workspace provided below. Label all edge weights.

Part B: Shortest Path Determination

Using the weighted adjacency matrix and your diagram, determine the **shortest path** and **total path weight** for each pair of nodes.

1. **Shortest Path from Node A to Node E:**

   Path Sequence: _____

   Total Weight: _____

2. **Shortest Path from Node C to Node H:**

   Path Sequence: _____

   Total Weight: _____

3. **Shortest Path from Node B to Node I:**

   Path Sequence: _____

   Total Weight: _____

4. **Shortest Path from Node A to Node G:**

   Path Sequence: _____

   Total Weight: _____

Part C: Travelling Salesperson Problem (TSP)

Find candidate closed tours that start and end at **Node A**, visit every node once, and minimize total weight.

Candidate Path Sequence 1: _____

Total Weight: _____

Candidate Path Sequence 2: _____

Total Weight: _____

Candidate Path Sequence 3: _____

Total Weight: _____

Complete Answer Key & Solutions

Problem 1 Solutions (Unweighted Graph)

Part B: Shortest Paths

1. **A to E:** $A \to B \to E$ (**2 hops**)
2. **C to H:** $C \to B \to E \to H$ or $C \to F \to E \to H$ (**3 hops**)
3. **B to I:** $B \to C \to F \to I$ or $B \to E \to H \to I$ (**3 hops**)
4. **A to G:** $A \to D \to G$ or $A \to J \to G$ (**2 hops**)

Part C: TSP Solution

- **Optimal Closed Tour:** $A \to B \to E \to H \to I \to F \to C \to D \to G \to J \to A$
- **Minimum Total Cost: 10 edges (hops)**
- **Strategy:** Since G_1 is a 3-regular unweighted graph with 10 vertices, any valid Hamiltonian cycle visits all 10 vertices with cost equal to exactly 10 edges.

Problem 2 Solutions (Weighted Graph)

Part B: Shortest Paths

1. **A to E:** $A \to B \to E$ – **Weight: 9** $(3 + 6)$
2. **C to H:** $C \to D \to G \to H$ – **Weight: 8** $(2 + 4 + 2)$
3. **B to I:** $B \to A \to J \to I$ – **Weight: 11** $(3 + 5 + 3)$
4. **A to G:** $A \to J \to G$ – **Weight: 11** $(5 + 6)$

Part C: TSP Solutions

- **Optimal Hamiltonian Cycle (Candidate 1):** $A \to B \to C \to D \to E \to F \to G \to H \to I \to J \to A$
  - **Total Weight: 37** $(3 + 4 + 2 + 5 + 3 + 4 + 2 + 6 + 3 + 5)$
- **Reverse Optimal Cycle (Candidate 2):** $A \to J \to I \to H \to G \to F \to E \to D \to C \to B \to A$
  - **Total Weight: 37**
- **Suboptimal Candidate Path (Candidate 3):** $A \to C \to B \to H \to I \to F \to E \to D \to G \to J \to A$
  - **Total Weight: 56** $(8 + 4 + 9 + 6 + 5 + 3 + 5 + 4 + 6 + 6)$